

Chapter 1: Factorial With Control

Introduction

Exploring a 2 by 2 Factorial with Control: A Case Study

Factorial experiments are widely used in scientific research for evaluating the interaction effects of multiple factors simultaneously. A 2×2 factorial design involves two factors, each at two levels, resulting in four treatment combinations. However, in many practical scenarios, researchers wish to compare these treatments not only among themselves but also against a standard or baseline treatment referred to as a control. In such cases, a 2×2 factorial design plus a control is adopted. There is an important difference between the traditional factorial design with a control and a 2 by 2 factorial design plus control. With the traditional factorial design with a control, we see all possible combinations of the factor levels including the control. With a factorial design plus a control, we only see possible combinations of treatments and their factor levels and control is used as a reference or considered the fifth treatment. We illustrate more of this concept using the example below.

Design Structure

Traditional 2×2 Factorial Design

In a traditional 2×2 factorial design, we have:

1. Factor A (e.g., Fertilizer type): Level A1 and A2
2. Factor B (e.g., Irrigation regime): Level B1 and B2 This gives us four treatment combinations:
3. A1B1
4. A1B2
5. A2B1

A2B2 All these combinations are part of the factorial structure. If a control treatment (e.g., no fertilizer and no irrigation) is included, it is often embedded into one of the treatment combinations (e.g., A1B1) or replicated in all combinations as a baseline within each cell.

2 × 2 Factorial Plus Control

In a factorial plus control design:

1. The control treatment is treated as a fifth, independent treatment, outside the factorial structure.
2. The four factorial combinations remain the same (A1B1, A1B2, A2B1, A2B2).
3. The control (C) is not associated with any levels of Factor A or B and is treated as its own level in the analysis.

Thus, there are 5 treatment groups in total: the four factorial combinations + 1 control.

A factorial design with control may be utilized in experiments where new drugs and methods are being tested for the first time or in new areas of study. In the case where treatments or methods are being tested for the first time, researchers may be interested in comparing the main effects of treatments with control, or in situations where they may be a significant interaction effect, compare each treatment with the control. In our example, we will compare each treatment with the control.

Example

Researchers are interested in evaluating the effect of two drugs on blood pressure. They would like to determine if a soluble or non-soluble form makes a difference. Another variable that affects performance of the drug is dosage (10 units and 15 units). Because this is a new study, they also chose a control group (No drug) to serve as reference. There were 3 replicates for each treatment combination and 3 people in the control group. A table showing the layout of the study is shown below.

Interpretation of Code

The `data.frame` function in R creates a data frame with variables Drug Type, Dosage, Description and Replicates. The `print` function then prints out the table.

R Code

	Drug_Type	Dosage	Description	Replicates
1	Soluble	10	Treatment A1B1	3
2	Soluble	15	Treatment A1B2	3
3	Non-soluble	10	Treatment A2B1	3
4	Non-soluble	15	Treatment A2B2	3
5	No Drug	0	Control	3

Data was simulated from the normal distribution with mean of zero and variance of 5. Also, data was simulated for the Control. The table above shows the treatments, Drug and Dosage, and the control. The function `set.seed()` is used to store our simulated data. This is very useful for reproducibility.

R Code

	Drug	Dosage	BloodPressure	Control
1	Soluble	10	128.1038	n
1.1	Soluble	10	125.1782	n
1.2	Soluble	10	128.8658	n
2	Non-soluble	10	136.3624	n
2.1	Non-soluble	10	131.8549	n
2.2	Non-soluble	10	129.1857	n
3	Soluble	15	127.9856	n
3.1	Soluble	15	125.6001	n
3.2	Soluble	15	124.2752	n
4	Non-soluble	15	136.5108	n
4.1	Non-soluble	15	131.0215	n
4.2	Non-soluble	15	124.2247	n
11	control	0	132.8952	y
21	control	0	138.8245	y
31	control	0	140.3308	y

SAS Code

```
data DrugData;
  input Drug $ Dosage BloodPressure Control $;
  datalines;
Soluble 10 128.1038 n
Soluble 10 125.1782 n
Soluble 10 128.8658 n
Non-soluble 10 136.3624 n
Non-soluble 10 131.8549 n
Non-soluble 10 129.1857 n
Soluble 15 127.9856 n
Soluble 15 125.6001 n
Soluble 15 124.2752 n
Non-soluble 15 136.5108 n
Non-soluble 15 131.0215 n
Non-soluble 15 124.2247 n
Control 0 132.8952 y
Control 0 138.8245 y
Control 0 140.3308 y
;
run;
```

```
proc print data=drugdata;
run;
```

Obs	Drug	Dosage	BloodPressure	Control
1	Soluble	10	128.104	n
2	Soluble	10	125.178	n
3	Soluble	10	128.866	n
4	Non-solu	10	136.362	n
5	Non-solu	10	131.855	n
6	Non-solu	10	129.186	n
7	Soluble	15	127.986	n
8	Soluble	15	125.600	n
9	Soluble	15	124.275	n
10	Non-solu	15	136.511	n
11	Non-solu	15	131.022	n
12	Non-solu	15	124.225	n
13	Control	0	132.895	y
14	Control	0	138.825	y
15	Control	0	140.331	y

Figure 1: Data for RCBD factorial with control

To set up a factorial design with control in R, one must set up the treatment combination as usual and only add the control as an additional factor. In our case, we first set up a 2×2 factorial design with the 2 levels in each factor using the `expand.grid` function and simulated the data. We then simulated the data for the control with the same standard deviation but different mean. We used the row-bind (`rbind()`) function to combine the two data frames.

A factorial with control design will require a variable that indicates whether a treatment is a control or not. Thus, in the final data frame, we see another variable called `Control` with entries “y” or “n”. Where “y” indicates the presence of control and “n” otherwise.

We equally obtain the same results with the corresponding SAS code with the `data` function which creates the data frame. The input function in SAS allows us to create the number of variables as demonstrated in the R code. In SAS, we use \$ sign after each variable to illustrate that it is categorical. We then use `proc print` function to print the data frame.

Statistical Challenge

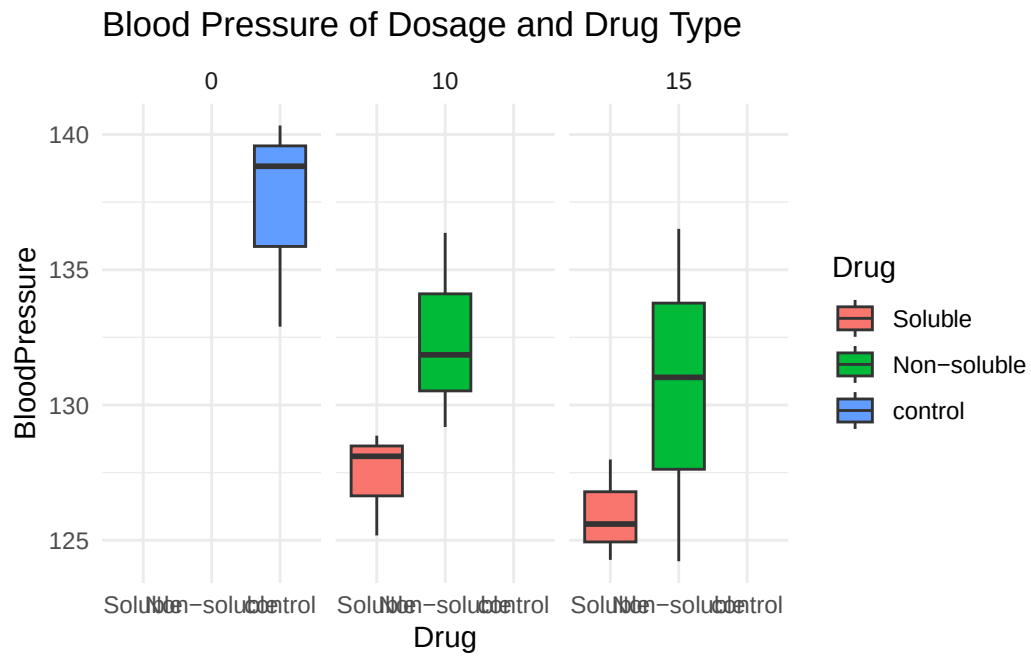
A lot of experiments, usually groundbreaking, take one form or shape like this experiment when a control is introduced as reference or baseline. The challenge is that a lot of researchers do not understand the difference between factorial design plus control and the traditional factorial design. Thereby analyzing such experiments as the latter. Such analysis is incorrect and only leads to wrong conclusions or inference. We demonstrate this point of view in the subsequent sections where we analyze data using the traditional factorial design and compare it to the factorial plus control design.

Exploratory Data Analysis

As with every data analysis, we conduct an exploratory data analysis to understand some characteristics of the data. We only present information that will be useful to the reader and researcher when using this design. We show the boxplot for the control and all other treatments. The boxplot shows the average blood pressure for control and all other treatment combinations using the `geom_boxplot` function in the library `ggplot2` package. The library `ggplot2` allows us to make some many nice and informative plots in R.

From the boxplot, we see those participants in the control have on average the highest blood pressure while the soluble drug at a dose of 15 units has the lowest average blood pressure.

R Code



SAS Code

```
proc sgplot data=DrugData;  
  vbox BloodPressure / category=Drug group=Dosage;  
  yaxis grid;  
run;
```

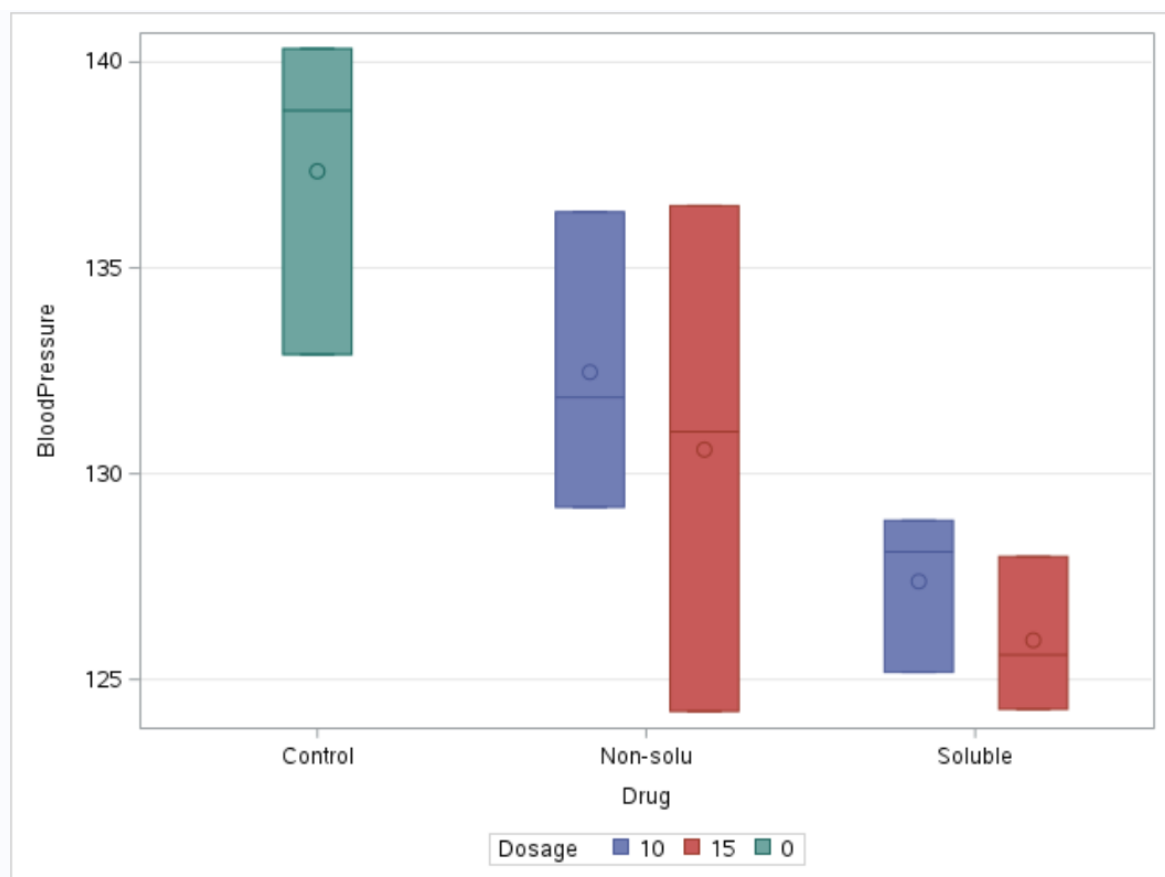


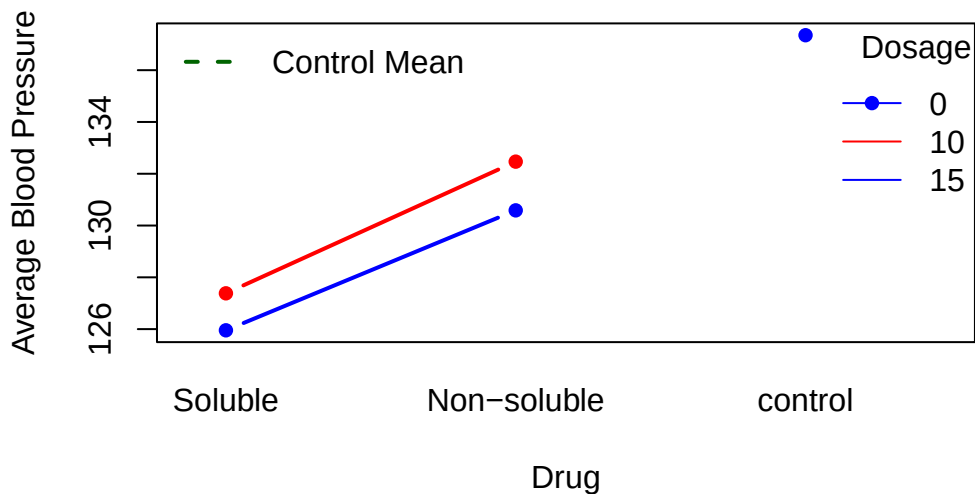
Figure 2: Boxplot of BloodPressure

Lastly, we also showed an interaction plot. From this plot, we see parallel lines which show there is no interaction between the two factors. Nevertheless, upon reading from a paper on a factorial design plus a control.

For a Completely Randomized Design, the control group is used as reference to all treatment combinations. For a Randomized Complete Block Design, one may need a control group in each block. For example, if we considered clinics as blocks, then there may be a control group in each clinic.

R Code

Interaction Plot of Drug and Dosage (with Control Reference)



SAS Code

```
data bp_plot;
  set bp_data;
  if Drug ^= "Control";
run;

/* Compute mean BloodPressure for Drug x Dosage */
proc means data=bp_plot noprint;
  class Drug Dosage;
```

```

var BloodPressure;
output out=means_plot mean=MeanBP;
run;

/* Remove _TYPE_ and _FREQ_ rows from PROC MEANS */
data means_plot_clean;
set means_plot;
if _TYPE_ = 3; /* Keeps only Drug x Dosage combos */
run;

/* Get Control mean */
proc means data=bp_data noprint;
where Drug = "Control";
var BloodPressure;
output out=control_mean mean=ControlMean;
run;

/* Add ControlMean to all rows for plotting */
data plot_ready;
if _n_ = 1 then set control_mean;
set means_plot_clean;
run;

/* Interaction Plot with Control Mean Line */
proc sgplot data=plot_ready;
series x=Drug y=MeanBP / group=Dosage lineattrs=(thickness=2) markers;
refline ControlMean / axis=y lineattrs=(color=darkgreen pattern=shortdash thickness=2);
yaxis label="Average Blood Pressure";
xaxis label="Drug";
keylegend / location=inside position=topright across=1;
title "Interaction Plot of Drug and Dosage (with Control Reference)";
run;

```

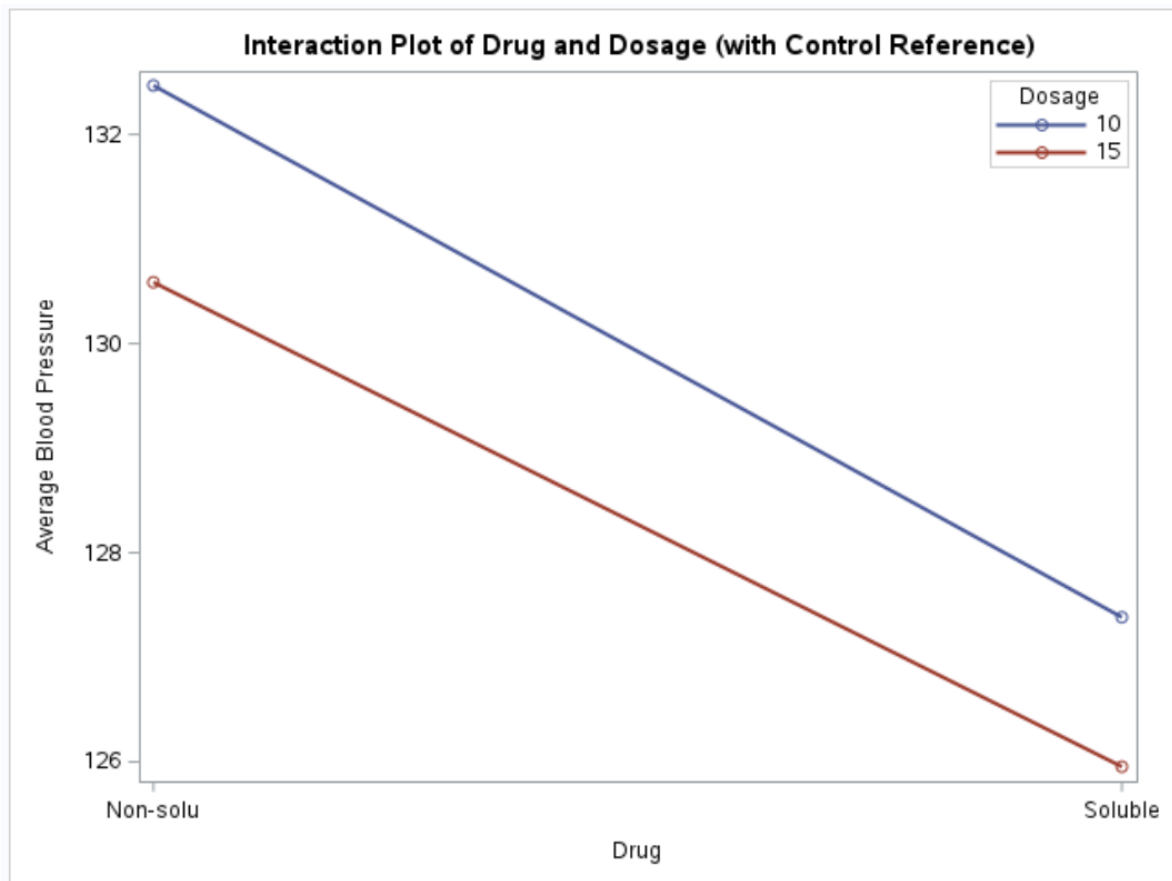


Figure 3: Interaction Plot of Drug and Dosage with Control as reference

Model

The linear model for a factorial design with control is given as;

$$y_{ijk} = \mu + \tau_{ik} + D_{jk} + control_k + \epsilon_{ijk}$$

where,

y_{ijk} is the blood pressure of the k th individual using the i th drug type at the j th dose

μ is the intercept

τ_{ik} is the effect from the i th drug type nested in the Control

D_{jk} is the effect from the j th dose level nested in the control

$control_k$ is the control

ϵ_{ijk} is the residual, $\epsilon_{ijk} \sim N(0, \sigma^2)$

ANOVA – General Structure

Consider a simple experiment with 2 factors, factors A and B, each with a and b levels respectively plus a control. Then the ANOVA table will look like the table below.

Data Analysis and Results in R

We first look at the ANOVA table when the data is analyzed as a 2 by 2 factorial design. Note that, analyzing the experiment in this manner is incorrect. Looking at the table below, we see that there is a significant effect of dosage, $p\text{-value} = 0.02112$. This is because the control is embedded in dosage and thus the estimated mean for dosage includes that of control. With this result, a researcher may believe that dosage of the drug was significant and may continue to explore which dosage led to the highest effect.

When we take a look at the correct model, we see that the $p\text{-value}$ for the control is statistically significant, $p\text{-value} = 0.00765$. Nevertheless, all other factors are not statistically significant including the interaction term. This is also supported by the parallel lines that we see with the interaction plot.

R Code

Analysis of Variance Table

Response: BloodPressure

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Dosage	2	171.683	85.841	5.8147	0.02112 *
Drug	1	70.818	70.818	4.7970	0.05332 .
Dosage:Drug	1	0.154	0.154	0.0104	0.92069
Residuals	10	147.629	14.763		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Dosage = 0:

Drug	emmean	SE	df	lower.CL	upper.CL
Soluble	nonEst	NA	NA	NA	NA
Non-soluble	nonEst	NA	NA	NA	NA
control	137	2.22	10	132	142

Dosage = 10:

Drug	emmean	SE	df	lower.CL	upper.CL
Soluble	127	2.22	10	122	132
Non-soluble	132	2.22	10	128	137
control	nonEst	NA	NA	NA	NA

Dosage = 15:

Drug	emmean	SE	df	lower.CL	upper.CL
Soluble	126	2.22	10	121	131
Non-soluble	131	2.22	10	126	136
control	nonEst	NA	NA	NA	NA

Confidence level used: 0.95

R Code

Analysis of Variance Table

Response: BloodPressure

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Control	1	163.461	163.461	11.0724	0.00765 **
Dosage	1	8.222	8.222	0.5569	0.47267
Drug	1	70.818	70.818	4.7970	0.05332 .

```
Dosage:Drug 1 0.154 0.154 0.0104 0.92069
Residuals 10 147.629 14.763
```

```
---
```

```
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Control = y, Dosage = 0:
```

Drug	emmean	SE	df	lower.CL	upper.CL
control	137	2.22	10	132	142

```
Control = n, Dosage = 10:
```

Drug	emmean	SE	df	lower.CL	upper.CL
Soluble	127	2.22	10	122	132
Non-soluble	132	2.22	10	128	137

```
Control = n, Dosage = 15:
```

Drug	emmean	SE	df	lower.CL	upper.CL
Soluble	126	2.22	10	121	131
Non-soluble	131	2.22	10	126	136

```
Confidence level used: 0.95
```

For a simple factorial plus a control like the example above, we seek to do these:

1. Determine whether there is a significant interaction between two treatments factors
2. Compare treatment to the control
3. Compare treatment to each other

We first show the Anova table for this model. The table shows all sources of variation, degrees of freedom, sum of squares, f-values and p-values. We see a significant difference between control and other treatments, $p\text{-value} = 0.00765$. We also see a marginal significance in effect for drug but no significant interaction effect between Drug and Dosage, $p\text{-values}$ 0.05332 and 0.92069 respectively. These values correspond to the interaction plot we saw earlier. Since there is no significant interaction effect, we can look at the main effects for Drug (soluble or non-soluble) to determine if there is any significant difference.

Lastly, we present the table for the estimated marginal means which shows the marginal means, standard errors and confidence intervals. From the ANOVA table, we saw that Drug was marginally significant. Therefore, we would write some contrast to determine if there is a significant difference between soluble and non-soluble drugs.

From the table below, there is no significant difference in effect between soluble and non-soluble drug and between non-soluble drug and control, $p\text{-values}$ are 0.1213 and 0.1304 respectively at 5% significance level. Nevertheless, there is a significant difference in effect between soluble drug and the control group, $p\text{-value} = 0.0072$.

R Code

Call:

```
lm(formula = BloodPressure ~ Control + Dosage + Drug + Drug *  
    Dosage, data = bp_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3609	-1.9414	0.4358	1.7576	5.9251

Coefficients: (5 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	125.954	2.218	56.779	6.97e-14 ***
Controly	11.397	3.137	3.633	0.00459 **
Dosage10	1.429	3.137	0.456	0.65848
Dosage15	NA	NA	NA	NA
DrugNon-soluble	4.632	3.137	1.477	0.17059
Drugcontrol	NA	NA	NA	NA
Dosage10:DrugNon-soluble	0.453	4.437	0.102	0.92069
Dosage15:DrugNon-soluble	NA	NA	NA	NA
Dosage10:Drugcontrol	NA	NA	NA	NA
Dosage15:Drugcontrol	NA	NA	NA	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.842 on 10 degrees of freedom

Multiple R-squared: 0.6217, Adjusted R-squared: 0.4704

F-statistic: 4.109 on 4 and 10 DF, p-value: 0.03182

Analysis of Variance Table

Response: BloodPressure

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Control	1	163.461	163.461	11.0724	0.00765 **
Dosage	1	8.222	8.222	0.5569	0.47267
Drug	1	70.818	70.818	4.7970	0.05332 .
Dosage:Drug	1	0.154	0.154	0.0104	0.92069
Residuals	10	147.629	14.763		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R Code

Control = n:

Drug	Dosage	emmean	SE	df	lower.CL	upper.CL
Soluble	10	127	2.22	10	122	132
Non-soluble	10	132	2.22	10	128	137
Soluble	15	126	2.22	10	121	131
Non-soluble	15	131	2.22	10	126	136

Control = y:

Drug	Dosage	emmean	SE	df	lower.CL	upper.CL
control	0	137	2.22	10	132	142

Confidence level used: 0.95

Drug	Dosage	Control	emmean	SE	df	lower.CL	upper.CL
Soluble	10	n	127	2.22	10	122	132
Non-soluble	10	n	132	2.22	10	128	137
Soluble	15	n	126	2.22	10	121	131
Non-soluble	15	n	131	2.22	10	126	136
control	0	y	137	2.22	10	132	142

Confidence level used: 0.95

contrast	estimate	SE	df	t.ratio	p.value
(Non-soluble Dosage10 n) - Soluble Dosage10 n	5.09	3.14	10	1.621	0.3599
Soluble Dosage15 n - Soluble Dosage10 n	-1.43	3.14	10	-0.456	0.9448
(Non-soluble Dosage15 n) - Soluble Dosage10 n	3.20	3.14	10	1.021	0.6874
control Dosage0 y - Soluble Dosage10 n	9.97	3.14	10	3.177	0.0327

P value adjustment: dunnett method for 4 tests

Control = n:

contrast	estimate	SE	df	t.ratio
Soluble Dosage10 - (Non-soluble Dosage10)	-5.09	3.14	10	-1.621
Soluble Dosage10 - Soluble Dosage15	1.43	3.14	10	0.456
Soluble Dosage10 - (Non-soluble Dosage15)	-3.20	3.14	10	-1.021
(Non-soluble Dosage10) - Soluble Dosage15	6.51	3.14	10	2.076
(Non-soluble Dosage10) - (Non-soluble Dosage15)	1.88	3.14	10	0.600
Soluble Dosage15 - (Non-soluble Dosage15)	-4.63	3.14	10	-1.477

p.value
0.4105

0.9670
 0.7415
 0.2251
 0.9298
 0.4850

```
Control = y:
contrast
(nothing)
p.value
NA
```

	estimate	SE	df	t.ratio
	nonEst	NA	NA	NA

P value adjustment: tukey method for varying family sizes

SAS Code

```
proc glm data=bp_data;
  class Control Drug Dosage;
  model BloodPressure = Control Drug(Control) Dosage(Control) Drug*Dosage(Control);
  lsmeans Drug(Control) Drug*Dosage(Control)/ pdiff=all adjust=dunnett;
run;
```

Estimated Marginal Means

```
proc glm data=bp_data;
  class Control Drug Dosage;
  model BloodPressure = Control Drug(Control) Dosage(Control) Drug*Dosage(Control);
  lsmeans Drug(Control) Drug*Dosage(Control)/ pdiff=all adjust=dunnett;
run;
```

Interpretation

In summary, the above example and results show that without any drug, blood pressure will be high on average. That is, there is a significant difference in effect between control and treatments. Nevertheless, we cannot tell based on the simulated data that a soluble drug may reduce blood pressure better than a non-soluble drug. Moreover, we cannot tell whether increasing the dosage from 10 units to 15 units will decrease blood pressure.

The GLM Procedure					
Dependent Variable: BloodPressure					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	242.6522374	60.6630593	4.11	0.0318
Error	10	147.6297543	14.7629754		
Corrected Total	14	390.2819917			

R-Square	Coeff Var	Root MSE	BloodPressure Mean
0.621736	2.938678	3.842262	130.7479

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Control	1	163.4599085	163.4599085	11.07	0.0076
Drug(Control)	1	70.8165243	70.8165243	4.80	0.0533
Dosage(Control)	1	8.2218752	8.2218752	0.56	0.4727
Drug*Dosage(Control)	1	0.1539294	0.1539294	0.01	0.9207

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Control	1	163.4599085	163.4599085	11.07	0.0076
Drug(Control)	1	70.8165243	70.8165243	4.80	0.0533
Dosage(Control)	1	8.2218752	8.2218752	0.56	0.4727
Drug*Dosage(Control)	1	0.1539294	0.1539294	0.01	0.9207

Figure 4: ANOVA Tables using Type I and Type III Sum of Squares

The GLM Procedure				
Least Squares Means				
Adjustment for Multiple Comparisons: Dunnett				
Drug	Dosage	Control	BloodPressure LSMEAN	H0:LSMean=Control Pr > t
Non-solu	10	n	132.467667	
Non-solu	15	n	130.585667	0.9302
Soluble	10	n	127.382600	0.3537
Soluble	15	n	125.953633	0.1831
Control	0	y	137.350167	0.3855

Figure 5: Table showing estimated marginal means

Conclusion

The above example illustrates the benefits of designing an experiment in this form. This design allowed researchers to answer some important questions which are usually present in studies that may be groundbreaking or there may not be enough research already done in that field.